

EE45: Intro to Foundations of ML

Mod 1 - Lec 2: Data Fitting Concepts Part Deux

Overview: [Module 1] Data Fitting Concepts

- This week
 - ▶ How to measure "fit" of a model to data?: norms, distances, inner products
 - ▶ Example: k -means clustering

References:

- [VMLS]: Chapters 1–3
- Topics: Vectors, Inner product, Linear functions, Regression model, Norm and distance

Boyd + VMLS

Content Summary

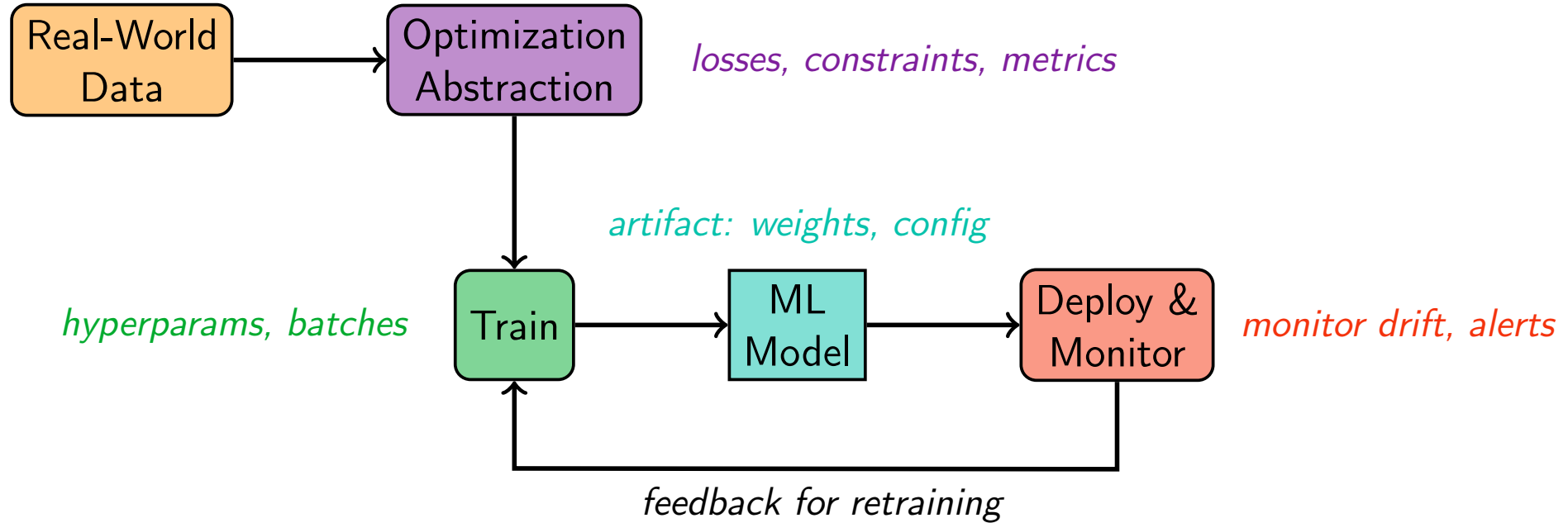
- Last time...
 - ▶ Formulate/abstract ML problems as an optimization problem
 - ▶ Similarity and distance measures as a key ingredient
 - ▶ Loss functions from distance measures such as Euclidean distance/norm, or absolute value, e.g.
 - ▶ Introduction to vector notation

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 - ▶ Similarity and distance measures as a key ingredient
 - ▶ Loss functions from distance measures such as Euclidean distance/norm, or absolute value, e.g.
 - ▶ Introduction to vector notation
- This time...
 - ▶ reinforce the vector notation via examples
 - ▶ Different Norms as loss fns how determines the geometry of ML problem ; soln.
k-means

ML workflow: Role of Linear Algebra & Optimization

logs, events, labels



Formulating a Learning Problem as Optimization

$$\mathcal{X} \subseteq \mathbb{R}^n, \quad \mathcal{Y} \subseteq \mathbb{R}$$

Model

A model is a $f_\theta : \mathcal{X} \rightarrow \mathcal{Y}$ recipe or rule that takes input data $x \in \mathcal{X}$ and produces a prediction $\hat{y} = f_\theta(x) \in \mathcal{Y}$.

- The quality of a model is measured by how well it fits the data. We can turn this into a number (an **error** or **loss**).
- We can then **optimize** that number.
- The loss is usually built from a **distance** (how far?) or a **similarity** (how aligned?).

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- The loss is usually built from a **distance** (how far?) or a **similarity** (how aligned?).
- Machines need to measure similarity, and distances are the key
 - ▶ Norms and distances turn subjective ideas (are these two cats alike?) into math: $\|x - y\|$.
 - ▶ Different norms capture different ideas of similarity (Euclidean vs Manhattan vs cosine).

Vector Notation

- To measure similarity we first need to represent data as vectors.
- Last time we saw the example of digits from MNIST represented as vectors based on their pixel values.
- Each data point with d features is just a point in $\mathbb{R}^d$.

E.g. housing prices.

feature vector

$\triangleright x = (x_1, x_2, \dots, x_d)$: $x_1 = \# \text{ beds}$, $x_2 = \text{sq ft.}$, $x_3 = \text{walkability}$

$\triangleright \theta = (\theta_1, \theta_2, \dots, \theta_d)$

model parameters.

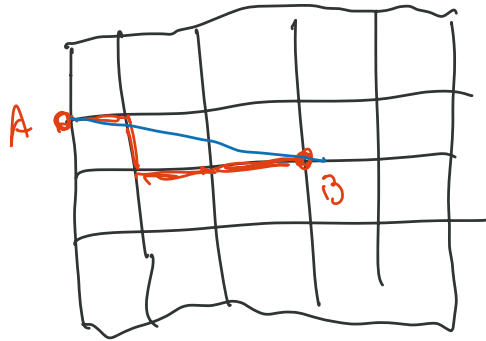
θ_2 : price per sq.ft., θ_1 : price per bed..

$$f_{\theta}(x) = \theta_1 x_1 + \theta_2 x_2 + \dots + \theta_d x_d$$

Length of a vector

- We now have a way of writing data as vectors.
- **Norm = measure of size/distance**: Norms let us quantify how big a vector is or how far apart two vectors are.
- Examples:

$$\bullet \|x\|_2 = \sqrt{x_1^2 + x_2^2 + \dots + x_d^2} \quad \text{Euclidean Norm}$$
$$\bullet \|x\|_1 = |x_1| + |x_2| + \dots + |x_d| \quad \text{Taxi cab Norm}$$



Norm and distance

$$\sqrt{\sum_{i=1}^n (x_i + y_i)^2} \leq \left(\sum_{i=1}^n x_i^2\right)^{1/2} + \left(\sum_{i=1}^n y_i^2\right)^{1/2} \quad x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}, y = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$$

- A norm is a function that measures the length of a vector.

- Euclidean norm (2-norm) is $\|x\| = \sqrt{\sum_{i=1}^n x_i^2}$

- properties: $\alpha \in \mathbb{R}$ is a scalar, $x, y \in \mathbb{R}^n$ are vectors

- ▶ homogeneity: $\|\alpha x\| = |\alpha| \|x\|$

- ▶ triangle inequality: $\|x+y\| \leq \|x\| + \|y\|$

- ▶ nonnegativity: $\|x\| \geq 0$

- ▶ definiteness: $\|x\| = 0$ if and only if $x = 0$ (iff)

- ▶ ~~triangle inequality:~~

$$\begin{aligned} \|\alpha x\| &= \sqrt{\sum_{i=1}^n (\alpha x_i)^2} \\ &= \sqrt{\sum_{i=1}^n \alpha^2 x_i^2} \\ &= \sqrt{\alpha^2 \sum_{i=1}^n x_i^2} \\ &= (\alpha^2)^{1/2} \left(\sum_{i=1}^n x_i^2 \right)^{1/2} \\ &= |\alpha| \|x\| \end{aligned}$$

Inner Products

$$\underline{1}_n = \underline{1} = (1, 1, \dots, 1)$$

- inner product (or dot product) $\langle x, y \rangle : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ of n -vectors x and y is

$$x^T y = [x_1 \dots x_n] \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$$

$$\langle x, y \rangle = x^T y = \sum_{i=1}^n x_i y_i$$

$$0_n = (0, 0, \dots, 0)$$

$$e_i = (0, \dots, 0, 1, 0, \dots)$$

satisfying:

► **Linearity:** for $a, b \in \mathbb{R}$: $\langle ax + by, z \rangle = a \langle x, z \rangle + b \langle y, z \rangle$

► **Symmetry:** $\langle x, y \rangle = \langle y, x \rangle$

► **Positivity:** $\langle x, x \rangle \geq 0 \rightarrow x_1^2 + x_2^2 + \dots + x_n^2 \geq 0$

► **Cauchy-Schwarz inequality:** $|\langle x, y \rangle| \leq \|x\| \|y\|$

► **Homogeneity:** $\langle \alpha x, y \rangle = \alpha \langle x, y \rangle$

• Examples: $e_i^T x = x_i$, $\underline{1}^T x = \sum_{i=1}^n x_i$

Examples

Matrix-Vector Multiplication as Inner Products

$d=4$ x, y , $f_{\theta}(x_i) = \theta_1 x_{i1} + \dots + \theta_d x_{id} = \theta^T x = x^T \theta$

$$\rightarrow \begin{bmatrix} x_{11} & x_{12} & x_{13} & x_{14} \\ x_{21} & x_{22} & x_{23} & x_{24} \\ x_{31} & x_{32} & x_{33} & x_{34} \end{bmatrix} \begin{bmatrix} \theta_1 \\ \theta_2 \\ \theta_3 \\ \theta_4 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

$$= \begin{bmatrix} \text{---} x_1^T \text{---} \\ \text{---} x_2^T \text{---} \\ \text{---} x_3^T \text{---} \end{bmatrix} \begin{bmatrix} \theta_1 \\ \vdots \\ \theta_4 \end{bmatrix} = \begin{bmatrix} x_1^T \theta \\ x_2^T \theta \\ x_3^T \theta \end{bmatrix} = \begin{bmatrix} \langle x_1, \theta \rangle \\ \langle x_2, \theta \rangle \\ \langle x_3, \theta \rangle \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

Inner Products Induce Norms

- Every inner product induces a norm: $\|x\| = \sqrt{\langle x, x \rangle}$
- In the "standard" inner product:

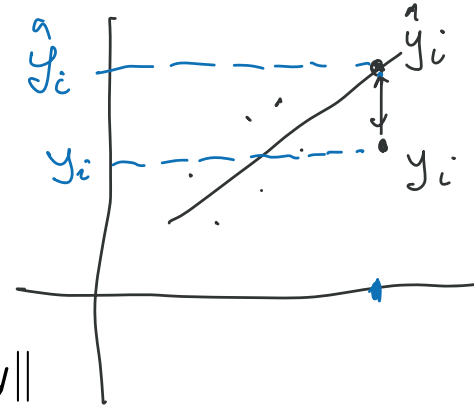
$$\|x\|_2 = (x^T x)^{1/2} = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2} = \sqrt{\langle x, x \rangle}$$

- To check: check that if we define the norm as $\|x\| = \langle x, x \rangle^{1/2}$ then all the norm properties hold (homogeneity, triangle inequality, etc.)

$$\|x\|_{2,M} \quad x^T M x \geq 0 \quad \begin{bmatrix} M \end{bmatrix} = M^T \quad \begin{bmatrix} 1 & 1 & \dots & 1 \\ m_1 & m_2 & \dots & m_d \\ 1 & 1 & \dots & 1 \end{bmatrix}^T = \begin{bmatrix} -m_1^T - \\ -m_2^T - \\ \vdots \\ -m_n^T - \end{bmatrix}$$

- To check: $\|x\|_M = \sqrt{x^T M x}$ for any symmetric positive definite matrix M is a norm.
- However, not every norm is induced by an inner product.

Why Norms?: Norms as Distances & Measures of Error



- Measuring error between prediction and truth:

$$\text{Error} = \|\hat{y} - y\|$$

- Example: residuals $r_i = y_i - \hat{y}_i$ and error $\|r\|_2 = \sqrt{\sum_{i=1}^n r_i^2}$
- Different norms $\Rightarrow$ different geometry of error

Norms

$$f(x_i) = \hat{y}_i, \quad y_i - \hat{y}_i = r_i$$

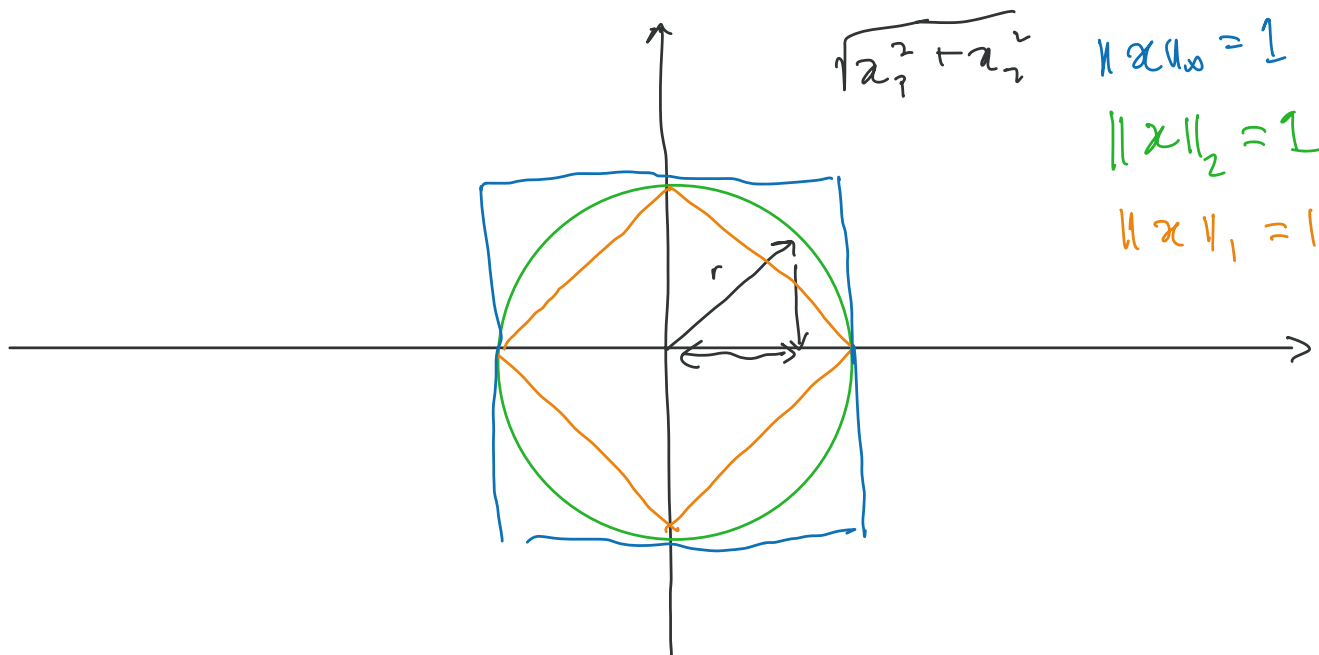
Define residual vector $r = (r_1, \dots, r_n)^T \in \mathbb{R}^n$.

$$\|r\|_p = \begin{cases} \left(\sum_{i=1}^n |r_i|^p \right)^{1/p}, & 1 \leq p < \infty \\ \max_i |r_i| & p = \infty \end{cases}$$

- ℓ_1 : $\|r\|_1 = \sum |r_i|$ "Taxi cab distance"
- ℓ_2 : $\|r\|_2 = \left(\sum |r_i|^2 \right)^{1/2}$
- ℓ_∞ : $\|r\|_\infty = \max_i |r_i|$ "Worst case error"

Choosing Norms

- $\|\cdot\|_2$: “mean fit” $\rightarrow$ sensitive to outliers, emphasizes large errors (squares them)
- $\|\cdot\|_1$: “median fit” $\rightarrow$ more robust.
- $\|\cdot\|_\infty$: “worst-case fit” $\rightarrow$ minimizes maximum deviation.



Error Growth

- ℓ_2 (squared error): Errors grow **quadratically**.

$$\|r\|_2^2 = \sum_i r_i^2$$

► Residual = 10 $\rightarrow$ contributes 100; Residual = 100 $\rightarrow$ contributes 10,000

- ℓ_1 (absolute error): Errors grow **linearly**.

$$\|r\|_1 = \sum_i |r_i|$$

► Residual = 10 $\rightarrow$ contributes 10; Residual = 100 $\rightarrow$ contributes 100

- ℓ_∞ : Errors grow like the **largest residual**.

$$\|r\|_\infty = \max_i |r_i|$$

Norms as Loss Functions

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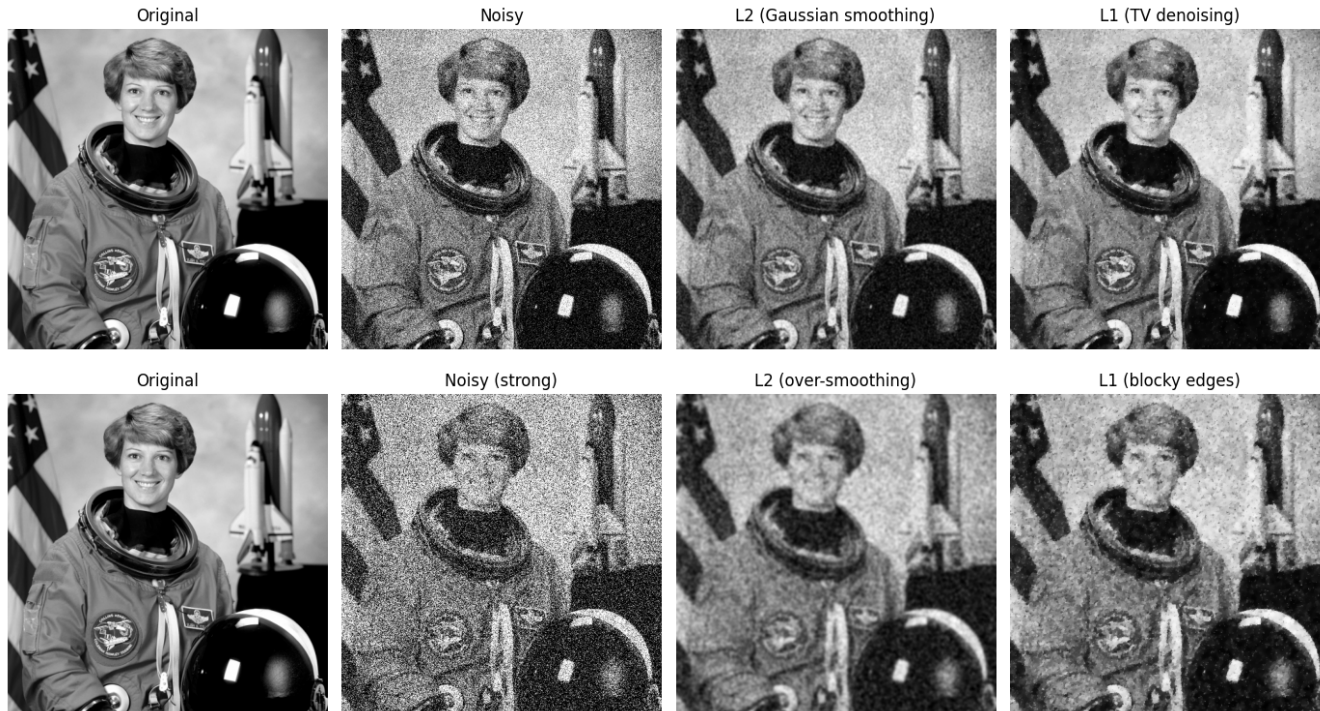
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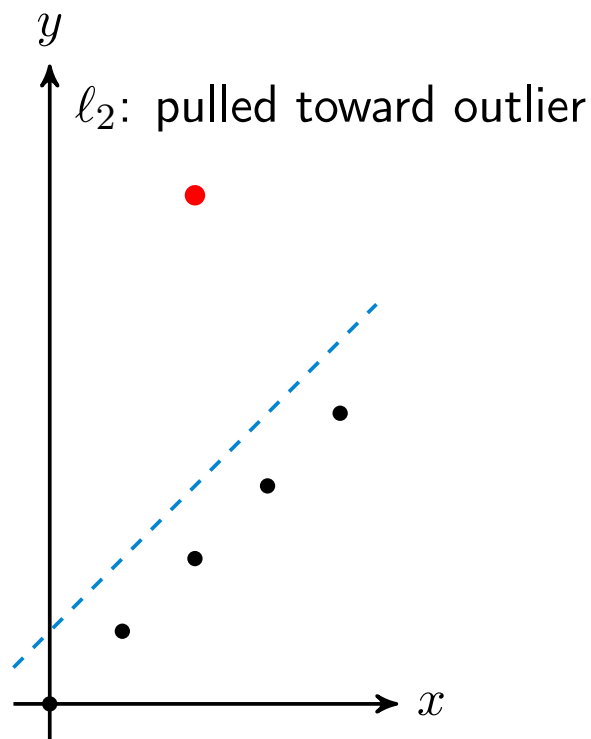
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 - ▶ Use case: Image reconstruction in medical imaging (MRI, CT) where data is undersampled.
 - ▶ ℓ_1 Encourages sparsity and robustness.

Example Image Reconstruction

- ℓ_2 produces smooth, averaged outputs.
- ℓ_1 produces blocky, median-like images.

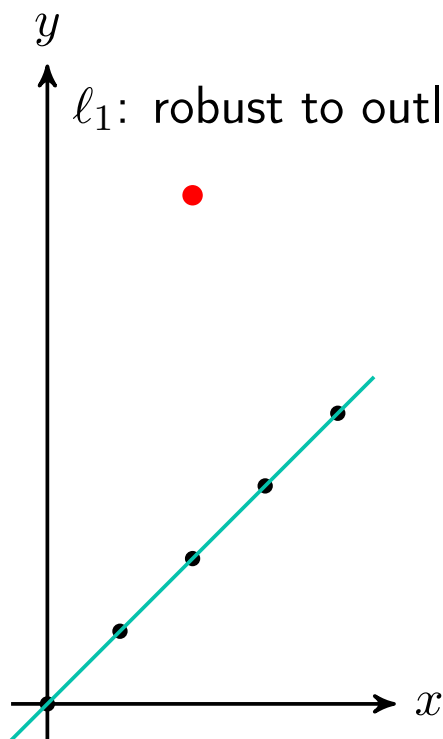
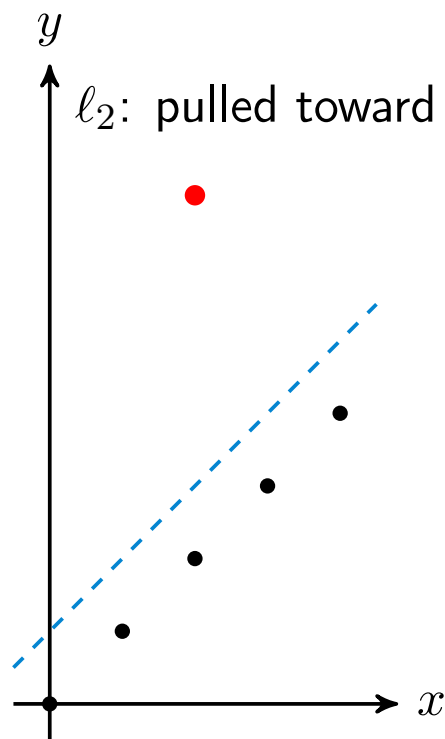


Fitting a line with one extreme outlier: ℓ_2 vs ℓ_1 vs ℓ_∞



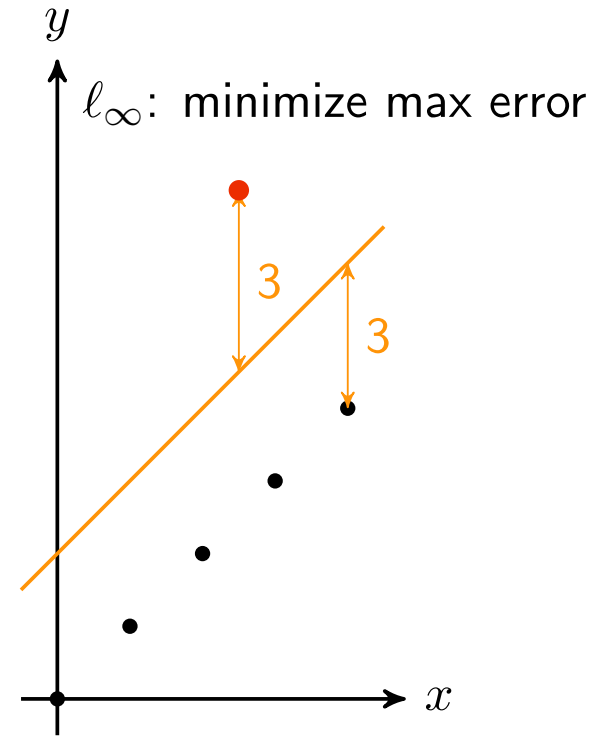
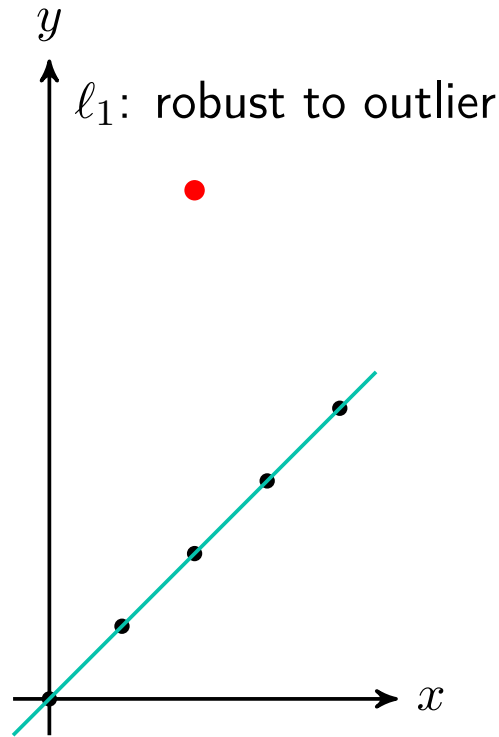
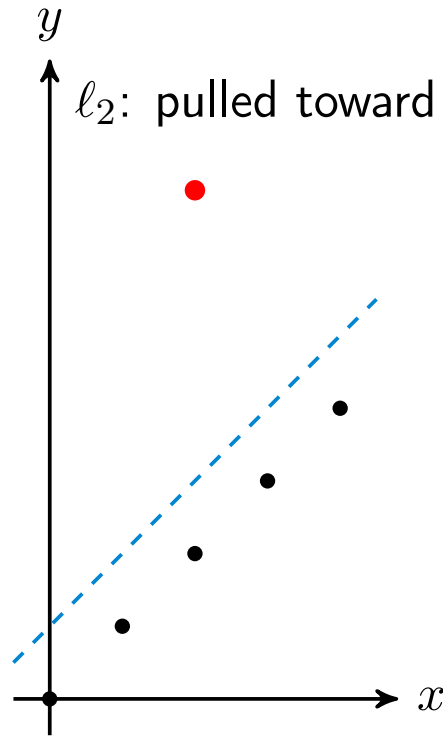
- ℓ_2 : Large errors dominate (quadratic), so the fit bends toward the outlier.
- ℓ_1 : Linear growth; outlier has limited pull, line stays with the bulk.
- ℓ_∞ : *minimizes the largest absolute residual*

Fitting a line with one extreme outlier: ℓ_2 vs ℓ_1 vs ℓ_∞



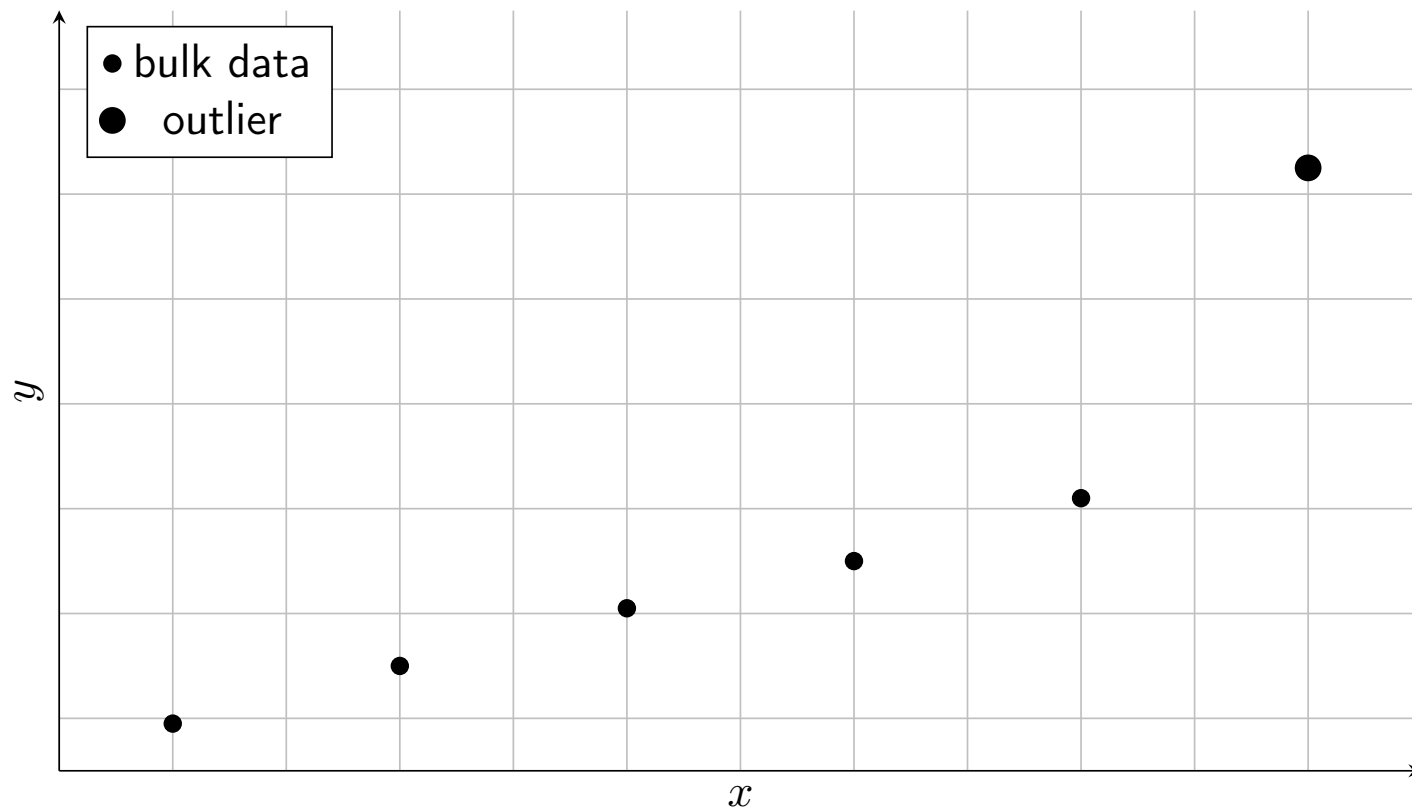
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Choosing Norm Impacts Model Fit: Let's apply the logic



Note: Without this outlier, ℓ_2 and ℓ_1 give nearly identical fits. The outlier is the stress test.

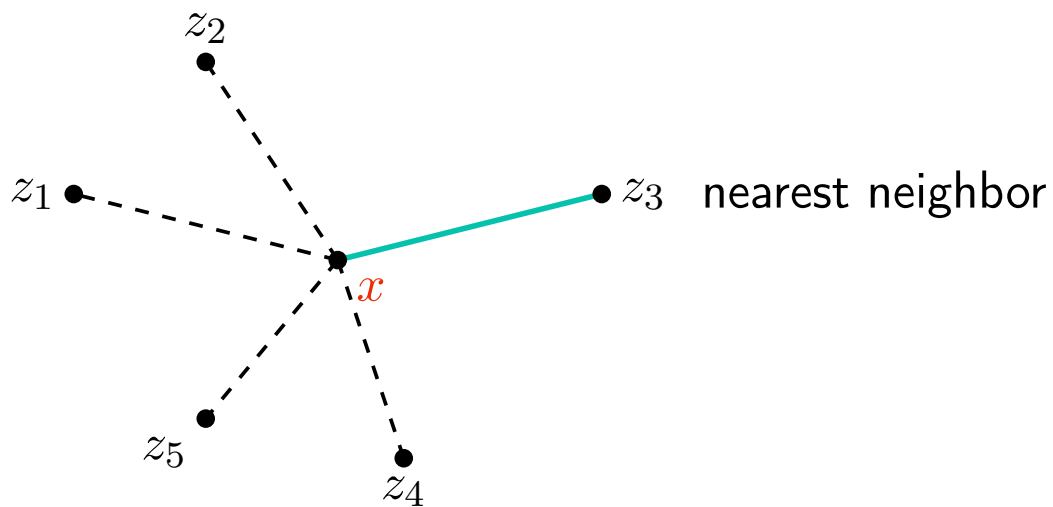
Why Distance Matters

- We saw norms as a way to define losses based on errors.
- Distances define what it means for points to be "close" or "similar."
- Nearest Neighbor and k -Means clustering rely entirely on this notion.
- Different distances $\Rightarrow$ different neighbors and clusters.

Feature distance and nearest neighbors

- if x, y are feature vectors for two entities, $\|x - y\|$ is the *feature distance*
- for vectors $z_1, \dots, z_m$, z_j is nearest neighbor of x if

$$\|x - z_j\| \leq \|x - z_i\|, \quad i = 1, \dots, m$$



Example: document dissimilarity

- 5 Wikipedia articles: 'Veterans Day', 'Memorial Day', 'Academy Awards', 'Golden Globe Awards', 'Super Bowl'
- word count histograms, dictionary of 4423 words (more setup details in sec 4.4.5 of VMLS)
- pairwise distances shown below:



	Veterans' day	Memorial day	Academy A.	Golden G.	Super Bowl
→ Veterans' day	0	0.095	0.130	0.153	0.170
Memorial day	0.095	0	0.122	0.147	0.164
Academy A.	0.130	0.122	0	0.108	0.164
Golden G.	0.153	0.147	0.108	0	0.181
Super Bowl	0.170	0.164	0.164	0.181	0

- we'll revisit this example later

Document Dissimilarity

- We may want to cluster documents into groups.
- e.g, awards, sports, holidays, etc.
- We can use k-means to cluster documents into k groups.

K-Means (Euclidean)

Goal. Partition data into k clusters by minimizing

$$J(\{\mu_j\}, z) = \sum_{i=1}^n \|x_i - \mu_{z_i}\|_2^2$$

Effect of Distance on K-Means

Euclidean (ℓ_2):

- Clusters are round (balls).
- Center = arithmetic mean.
- Sensitive to scale of features.

Manhattan (ℓ_1):

- Clusters look diamond-shaped.
- Center = componentwise median.
- More robust to outliers.

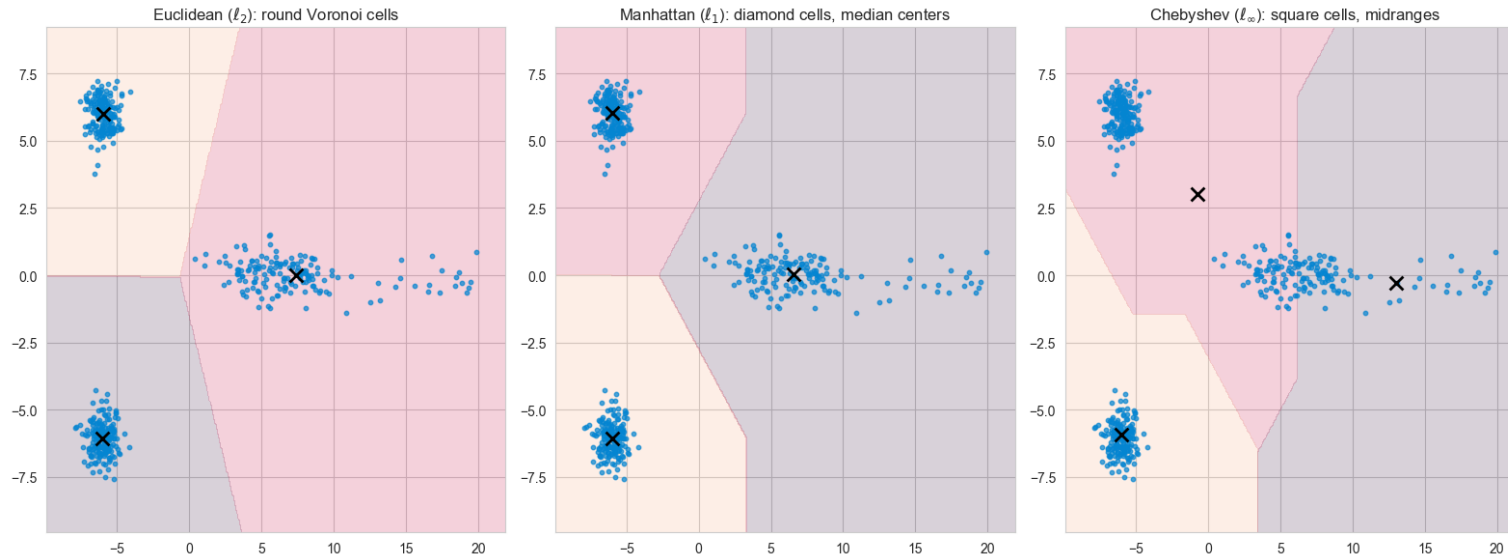
Cosine similarity:

- Clusters by angle, not magnitude.
- Center = normalized mean direction.
- Common for text or embeddings.

General lesson:

- Choice of distance = inductive bias.
- Different geometries $\Rightarrow$ different clusters.

K-Means with Different Distances



- ℓ_2 : round clusters, center on the elongated cluster is pulled toward the outliers (mean).
- ℓ_1 : diamond clusters, center sits closer to the componentwise median, resisting the long tail.
- ℓ_∞ : angular clusters, center sits at the midrange (avg of min/max per coordinate), so it shifts differently again.

k-NN vs k-Means

k-Nearest Neighbors (k-NN)

- Supervised learning algorithm
- Used for classification (and regression)
- Predicts label of a new point by majority vote of its k nearest neighbors
- Relies on a distance metric (Euclidean, Manhattan, etc.)
- No training phase beyond storing the data

k-Means Clustering

- Unsupervised learning algorithm
- Used for clustering unlabeled data
- Iteratively assigns points to k clusters and updates cluster centers
- Objective: minimize within-cluster variance
- Training = alternating assignment and update until convergence

Both use k and a distance metric, but serve very different purposes.

Document Dissimilarity: k-NN vs k-Means

Example corpus:

Veterans Day, Memorial Day, Academy Awards, Golden Globes, Super Bowl

k-Means (unsupervised)

- Input: pairwise document distances (from word-count histograms).
- Goal: cluster into $k = 3$ groups.
- Outcome:
 - ▶ Veterans Day, Memorial Day → Holidays
 - ▶ Academy Awards, Golden Globes → Awards
 - ▶ Super Bowl → Sports
- No labels required.

k-NN (supervised)

- Assume we already have labels: Holiday, Award Show, Sports.
- New document: “Labor Day.”
- Compute distances to all 5 documents.
- Nearest neighbors (Veterans Day, Memorial Day).
- Majority label = Holiday $\Rightarrow$ classify as Holiday.

Same distance representation, two different uses.

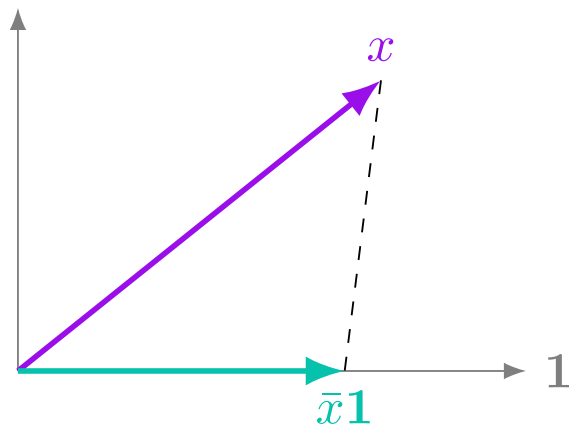
(If time permits) Standardizing Data

Example: Mean as Inner Product

- The mean is the “balance point” (center of mass).
- The sample mean is

$$\mu = \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i = \frac{\langle x, \mathbf{1} \rangle}{\langle \mathbf{1}, \mathbf{1} \rangle}$$

- Geometrically: $\bar{x} \cdot \mathbf{1}$ is the **projection** of x onto the "span" of $\mathbf{1}$
- Intuition: the mean is the "best constant approximation" of the data

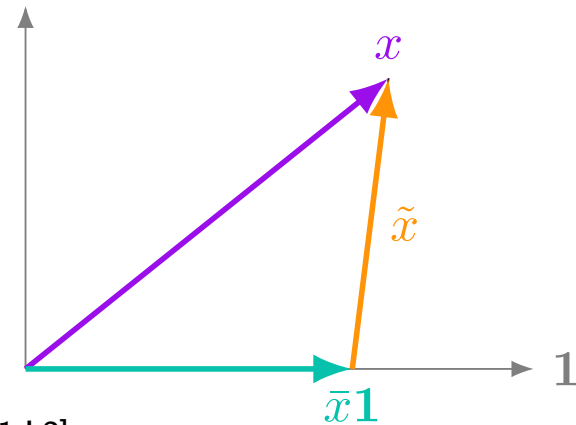


Centering Feature Vectors

Given $x \in \mathbb{R}^n$ and $\mathbf{1} = (1, 1, \dots, 1)^\top \in \mathbb{R}^n$:

$$\tilde{x} = x - \bar{x} \mathbf{1} = x - \frac{1}{n} \mathbf{1}^\top x \mathbf{1} = x - \frac{1}{n} \mathbf{1} \mathbf{1}^\top x = \left(I - \frac{1}{n} \mathbf{1} \mathbf{1}^\top \right) x$$

- $\mathbf{1} \mathbf{1}^\top \in \mathbb{R}^{n \times n}$ is the all-ones matrix.
- $H = I - \frac{1}{n} \mathbf{1} \mathbf{1}^\top$ is the **centering matrix**.
- H is symmetric and idempotent ($H^2 = H$): an **orthogonal projection**.
- Geometrically: projects x onto the subspace orthogonal to $\mathbf{1}$.



Example: Variance and Standard Deviation

- The centered data vector is $x - \mu \mathbf{1}$
- The standard deviation is the “typical distance” of points from that balance point.
 - ▶ Small SD $\rightarrow$ data clustered tightly around the mean.
 - ▶ Large SD $\rightarrow$ data spread out.
- If you pick a random data point, the standard deviation is roughly the size of the “typical error” you’d make if you always guessed the mean.
- Think of the mean as a reference position. The standard deviation is like the moment of inertia—how far mass is spread out from the center. Same “center,” but very different spread.
- The variance is the **scaled average squared deviation**

Example: Variance and Standard Deviation

- The centered data vector is $\tilde{x} = x - \bar{x}\mathbf{1} = Hx$, where $H = I - \frac{1}{n}\mathbf{1}\mathbf{1}^\top$ is the centering matrix.
- Its squared norm gives the total squared deviation:
- The variance is the **scaled average squared deviation**:
- Standard deviation (= avg length of the projected (centered) vector):

Standardizing Data

Given data $x = (x_1, \dots, x_n)$ with mean $\bar{x}$ and std. deviation s , the **standardized data** (z-scores) are

Properties

- Centered: $\text{mean}(z) = 0$
- Scaled: $\text{sd}(z) = 1$
- Interpretable in **standard deviation units**.
- Comparable across different variables

Standardizing Data (z-scores)

Given $x \in \mathbb{R}^n$, mean $\bar{x}$, and centering matrix the **centered data** is

The **standard deviation** is

Standardized data (z-scores):

Why (and When Not) to Standardize Data

- **Why standardize:**
 - ▶ **Comparability across features** Different units (kg, \$, counts) → put in common **std dev units**.
 - ▶ **Geometry matters** kNN, SVM, PCA use distances/dot products; large-scale features dominate otherwise.
 - ▶ **Optimization stability** Gradient descent converges faster; loss surface becomes “rounder.”
 - ▶ **Interpretability** Coefficients are comparable across features.

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- ▶ **Geometry matters** kNN, SVM, PCA use distances/dot products; large-scale features dominate otherwise.
- ▶ **Optimization stability** Gradient descent converges faster; loss surface becomes “rounder.”
- ▶ **Interpretability** Coefficients are comparable across features.

- **When **not** to standardize:**

- ▶ **Categorical features** encoded as one-hot or integers → standardizing breaks their meaning.
- ▶ **Features with physical meaning in the model** e.g. age in years, probability values, counts with direct interpretability.
- ▶ **Tree-based models** (decision trees, random forests, gradient boosting) → splits are order-based, not distance-based; scaling irrelevant.